Task-5

Titanic Dataset – Exploratory Data Analysis

1. Introduction

Exploratory Data Analysis (EDA) is an important step in data analysis used to understand the structure of a dataset, identify patterns, detect missing values and discover relationships between variables.

This project performs EDA on the **Titanic dataset** using Python, Pandas, Matplotlib and Seaborn. The analysis focuses on passenger characteristics and factors associated with survival.

2. Objective

The main objectives of this analysis are:

- Understand the structure and characteristics of the Titanic dataset.
- Identify missing and duplicate values.
- Explore numerical and categorical variables.
- Analyze relationships between passenger characteristics and survival.
- Identify important patterns and insights using visualizations and statistical summaries.

3. Dataset Overview

The dataset contains **891 passenger records and 12 original columns**.

Important variables include:

- **Survived** – Survival status (0 = No, 1 = Yes)
- **Pclass** – Passenger class
- **Sex** – Passenger gender
- **Age** – Passenger age
- **SibSp** – Number of siblings/spouses aboard
- **Parch** – Number of parents/children aboard
- **Fare** – Passenger fare
- **Embarked** – Port of embarkation
- **Cabin** – Cabin information

4. Data Cleaning

The dataset was checked for missing values and duplicate records.

The main missing values were found in:

- **Age:** 177 missing values
- **Cabin:** 687 missing values
- **Embarked:** 2 missing values

No duplicate records were found.

The following cleaning steps were performed:

- Missing **Age** values were replaced with the median age.
- Missing **Embarked** values were replaced with the most frequent value.
- The **Cabin** column was converted into a new `CabinKnown` feature indicating whether cabin information was available.
- The original Cabin column was removed after creating the new feature.

After cleaning, there were **no missing values** in the analysis dataset.

5. Exploratory Data Analysis

Survival Distribution

The dataset contains:

- **549 passengers who did not survive**
- **342 passengers who survived**

This shows that the number of non-survivors was higher than the number of survivors.

Passenger Gender

There were:

- **577 male passengers**
- **314 female passengers**

The survival analysis showed that female passengers had better survival outcomes compared with male passengers.

Passenger Class

The passenger distribution was:

- **1st Class:** 216 passengers
- **2nd Class:** 184 passengers
- **3rd Class:** 491 passengers

Third-class passengers formed the largest group and also had the highest number of non-survivors.

6. Numerical Variable Analysis

Age Distribution

Most passengers were approximately between **20 and 40 years old**. There were fewer very young and elderly passengers.

The boxplot showed that most ages were concentrated within the central range, with a few older passengers appearing as potential outliers.

Fare Distribution

Most passengers paid relatively low fares, while a small number paid considerably higher fares.

The fare distribution was **right-skewed**, and the boxplot identified several high-fare values as potential outliers.

7. Relationship Analysis

Survival by Gender

The analysis showed a clear difference in survival outcomes between genders. Female passengers had a much higher proportion of survivors, while male passengers had considerably more non-survivors.

Survival by Passenger Class

Passenger class was also strongly associated with survival. First-class passengers had better survival outcomes, while third-class passengers experienced the highest number of non-survivals.

Age vs Survival

The scatterplot showed passengers of different ages in both survival groups. Therefore, age alone did not show a strong relationship with survival.

Age vs Fare by Survival

Most passengers had relatively low fares across different age groups. A small number of passengers paid much higher fares, and these higher-fare passengers were more commonly associated with the survival group.

8. Correlation Analysis

A correlation heatmap was used to examine relationships between numerical variables.

The analysis showed a **negative relationship between Pclass and Survived**. Since Pclass is coded as 1 for first class and 3 for third class, this indicates better survival among passengers in better passenger classes.

Other numerical variables showed relatively weaker relationships with survival.

9. Pairplot Analysis

The pairplot was used to visualize relationships among numerical variables including:

- Survived
- Pclass
- Age
- SibSp
- Parch
- Fare

The visualization helped identify patterns between the variables and showed that passenger class and fare had noticeable relationships with survival.

10. Key Findings

The major findings from the EDA are:

1. The Titanic dataset contains **891 passengers**.
2. **342 passengers survived**, while **549 did not survive**.
3. Female passengers had better survival outcomes than male passengers.
4. First-class passengers had better survival outcomes than lower-class passengers.
5. Third-class passengers had the highest number of non-survivors.
6. Most passengers were between approximately 20 and 40 years old.
7. Fare values were highly right-skewed.
8. Higher fares were more commonly associated with surviving passengers.
9. Age alone did not show a strong relationship with survival.
10. Passenger class was one of the more noticeable numerical factors associated with survival.

11. Conclusion

The exploratory analysis of the Titanic dataset revealed several important patterns related to passenger survival. **Gender, passenger class and fare** showed noticeable associations with survival, while age alone showed a weaker relationship.EDA provided a better understanding of the dataset and helped identify important variables and patterns that could be considered in further statistical analysis or predictive modeling.